

## Topic 4 — Linear Regression, Correlation, and Logistic Regression

## Linear Regression: Motivation

A **linear regression** fits a straight line to observed data to demonstrate a linear relationship between the variables, and it makes predictions on new data.

### Example

Exam marks vs. revision hours: this can be represented by a scatter plot, with a fitted line drawn through the data.

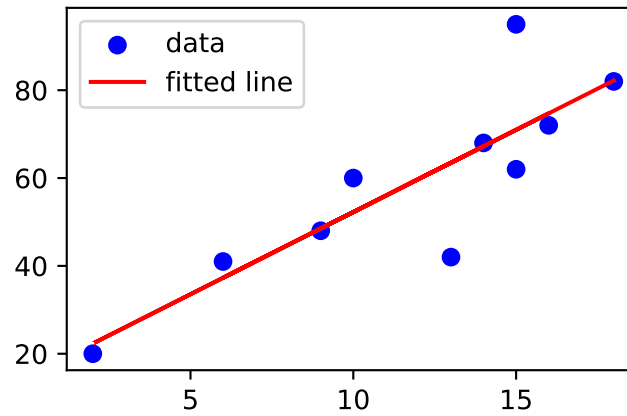


Figure 1

# The Method of Least Squares

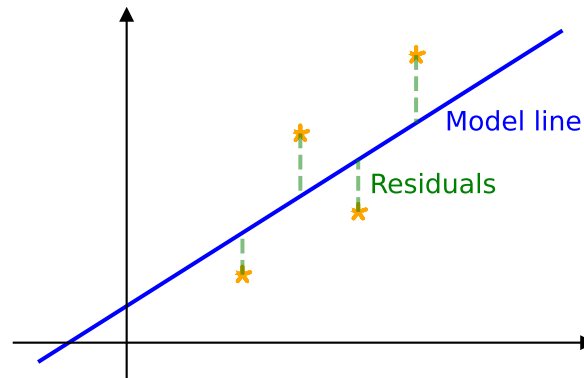


Figure 2

Assuming data pairs  $(x_1, y_1), \dots, (x_n, y_n)$ , the model line takes the form  $\mathbb{E}(y) = \beta_0 + \beta_1 x$ , where  $y$  is the **dependent (response) variable** and  $x$  is the **independent (explanatory/predictor) variable**. The variables are related by

$$y = \beta_0 + \beta_1 x + \epsilon,$$

where  $\epsilon$  is the error. We look for the line minimising the sum of squared residuals

$$f(\beta_0, \beta_1) = \sum_{i=1}^n (\beta_0 + \beta_1 x_i - y_i)^2.$$

## Least Squares Estimates

### Definition

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n x_i y_i - n\bar{x}\bar{y}}{\sum_{i=1}^n x_i^2 - n\bar{x}^2} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} = \frac{S_{xy}}{S_{xx}};$$

and

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}.$$

Note that

$$S_{yy} = \sum_{i=1}^n (y_i - \bar{y})^2 = \sum_{i=1}^n y_i^2 - n\bar{y}^2.$$

The least squares model is therefore

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x.$$

## Correlation

### Definition

The strength of a linear relationship between the variables can be measured by the **Pearson correlation coefficient**:

$$r = \frac{S_{xy}}{\sqrt{S_{xx}S_{yy}}}.$$

### Remark

$r$  can take values between  $-1$  and  $1$ , where  $-1$  and  $1$  represent a perfect linear relationship, and  $r = 0$  means no linear relationship. A positive  $r$  denotes a positive correlation, a negative  $r$  a negative correlation. As a rule of thumb:

$$\begin{array}{ll} |r| > 0.7 & \text{Strong correlation} \\ 0.7 \geq |r| > 0.4 & \text{Moderate correlation} \\ |r| \leq 0.4 & \text{Weak correlation} \end{array}$$

## Example: Revision Hours vs. Exam Marks

### Example

$$\bar{x} = 11.8, \bar{y} = 59.$$

|                                  |       |       |       |      |       |       |       |      |      |       |       |
|----------------------------------|-------|-------|-------|------|-------|-------|-------|------|------|-------|-------|
| Revision<br>Hours<br>( $x_i$ )   | 18    | 2     | 13    | 14   | 6     | 15    | 16    | 9    | 10   | 15    | Sum   |
| Mark<br>( $y_i$ )                | 82    | 20    | 42    | 68   | 41    | 95    | 72    | 48   | 60   | 62    |       |
| $x_i - \bar{x}$                  | 6.2   | -9.8  | 1.2   | 2.2  | -5.8  | 3.2   | 4.2   | -2.8 | -1.8 | 3.2   | 0     |
| $y_i - \bar{y}$                  | 23    | -39   | -17   | 9    | -18   | 36    | 13    | -11  | 1    | 3     | 0     |
| $(x_i - \bar{x})(y_i - \bar{y})$ | 142.6 | 382.2 | -20.4 | 19.8 | 104.4 | 115.2 | 54.6  | 30.8 | -1.8 | 9.6   | 837   |
| $(x_i - \bar{x})^2$              | 38.44 | 96.04 | 1.44  | 4.84 | 33.64 | 10.24 | 17.64 | 7.84 | 3.24 | 10.24 | 223.6 |
| $(y_i - \bar{y})^2$              | 529   | 1521  | 289   | 81   | 324   | 1296  | 169   | 121  | 1    | 9     | 4340  |

### Example

$$\hat{\beta}_1 = \frac{837}{223.6} = 3.743, \quad \hat{\beta}_0 = 59 - 3.743 \times 11.8 = 14.829.$$

Regression line:  $\hat{y} = 3.743x + 14.829$ .

$$r = \frac{837}{\sqrt{223.6}\sqrt{4340}} = 0.85,$$

indicating a strong positive correlation.  $r^2 = 0.723$ : 72.3% of the variance of the marks is explained by revision. The regression model fits the data well.

### Example

For 11 hours of revision:

$$\hat{y} = 3.743 \times 11 + 14.829 \approx 56.$$

Common sense should be used when predicting: we cannot predict outside the possible range of  $x$  values (e.g. -5 hours of revision makes no sense).

## Multiple Linear Regression

This extends the simple linear regression model  $y = \beta_0 + \beta_1 x + \epsilon$  to several independent variables  $x_1, \dots, x_n$  and a single dependent variable  $y$ :

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n + \epsilon.$$

We concentrate on  $n = 2$  independent variables:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \epsilon.$$

**Example.**  $x_1$  — hours of revision,  $x_2$  — number of attended lectures,  $y$  — mark.

Coding the variables:

$$v_i = y_i - \bar{y}, \quad u_{1i} = x_{1i} - \bar{x}_1, \quad u_{2i} = x_{2i} - \bar{x}_2,$$

we write the model as  $v = \beta'_0 + \beta_1 u_1 + \beta_2 u_2 + e$ , and minimise

$$Q = \sum_{i=1}^n (v_i - \beta'_0 - \beta_1 u_{1i} - \beta_2 u_{2i})^2.$$

This leads to the **normal equations**. Using

$$S_{pq} = \sum_{i=1}^n (x_{pi} - \bar{x}_p)(x_{qi} - \bar{x}_q), \quad S_{0p} = \sum_{i=1}^n (x_{pi} - \bar{x}_p)(y_i - \bar{y}), \quad p, q = 1, 2,$$

they become

$$\hat{\beta}_1 S_{11} + \hat{\beta}_2 S_{12} = S_{01}, \quad \hat{\beta}_1 S_{12} + \hat{\beta}_2 S_{22} = S_{02}.$$

With  $D = S_{11}S_{22} - S_{12}^2$ ,

$$\hat{\beta}_1 = \frac{S_{22}S_{01} - S_{12}S_{02}}{D}, \quad \hat{\beta}_2 = \frac{S_{11}S_{02} - S_{12}S_{01}}{D}, \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}_1 - \hat{\beta}_2 \bar{x}_2,$$

giving the model  $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2$ .

## Logistic Regression: Motivation

The main use of logistic regression is to predict a binary outcome from a linear combination of independent variables — e.g. predicting whether a student passes an exam from hours of revision. Here the data is binary, taking only values 0 and 1 (formally, the dependent variable  $Y \sim \text{Bin}(1, p)$ ), and we want to predict  $p$ .

## Example: Pass/Fail vs. Revision Hours

### Example

A student spent various numbers of hours revising, and we record whether they passed (1) or failed (0):

| Hours | 10 | 11 | 12 | 13 | 14 | 17 | 18 | 19 | 20 |
|-------|----|----|----|----|----|----|----|----|----|
| P/F   | 0  | 0  | 0  | 1  | 0  | 1  | 1  | 0  | 1  |

A scatter plot of this data (points at  $y = 0$  or  $y = 1$  for each hour value) is naturally fitted with an S-shaped (sigmoid) curve rising from near 0 for low hours to near 1 for high hours.

## The Logistic (Sigmoid) Model

### Definition

The logistic regression model chooses a fitting curve of the form

$$f(x) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x)}}, \quad (1 + e^{-(\beta_0 + \beta_1 x)}) > 0,$$

where  $\beta_0, \beta_1$  are fitting parameters estimated from the data. Writing  $y = \beta_0 + \beta_1 x$  and  $g(y) = \frac{1}{1 + e^{-y}} = \frac{e^y}{1 + e^y}$ , we have  $f(x) = g(\beta_0 + \beta_1 x)$ .

Always  $0 < f(x) < 1$ . If  $\beta_1 > 0$ , then  $\lim_{x \rightarrow +\infty} f(x) = 1$  and  $\lim_{x \rightarrow -\infty} f(x) = 0$ . Indeed,

$$\frac{1}{1 + e^{-\beta_1 x - \beta_0}} = \frac{1}{1 + \frac{e^{-\beta_0}}{e^{\beta_1 x}}} \rightarrow \frac{1}{1 + 0} = 1 \quad (x \rightarrow +\infty, \beta_1 > 0),$$

$$\frac{1}{1 + e^{-\beta_1 x - \beta_0}} \rightarrow 0 \quad (x \rightarrow -\infty).$$

Also,  $f(x)$  is increasing.

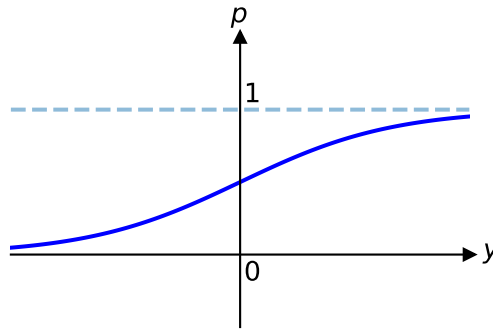


Figure 3

## Deriving the Logit Function

We use this curve for prediction:  $g(y) = p \in [0, 1]$  is the probability (e.g. of passing). Solving for  $y$ :

$$g(y) = \frac{e^y}{1 + e^y} = p \implies e^y = p + pe^y \implies e^y(1 - p) = p \implies e^y = \frac{p}{1 - p} \implies y = \ln \frac{p}{1 - p}.$$

### Definition

The quantity  $\frac{p}{1 - p}$  is called the **odds**.

### Definition

The natural logarithm of the odds is called the **logit function**:

$$\text{logit}(p) = \ln \frac{p}{1 - p}, \quad 0 < p < 1.$$

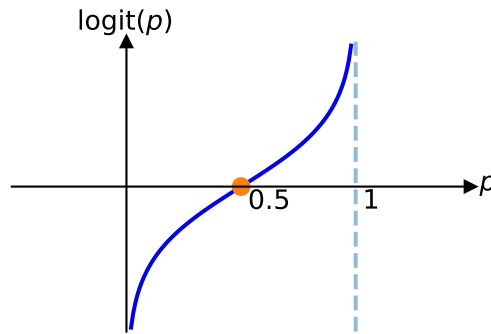


Figure 4

So  $g(y) = p$  means  $y = \text{logit}(p)$ : the logit is the inverse function of the logistic function

$$g(y) = \frac{1}{1 + e^{-y}} = \frac{e^y}{1 + e^y}.$$

## Several Predictor Variables

If the data depends on several predictor variables  $x_1, x_2, \dots, x_n$  (e.g.  $x_1$  = hours of revision,  $x_2$  = cups of coffee, ...), we still consider

$$f(x_1, \dots, x_n) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n)}},$$

with  $y = \beta_0 + \beta_1 x_1 + \dots + \beta_n x_n$ , so that  $f(x_1, \dots, x_n) = g(y) = p \in (0, 1)$ , and

$$\hat{\beta}_0 + \hat{\beta}_1 x_1 + \dots + \hat{\beta}_n x_n = y = \text{logit}(p),$$

where  $\hat{\phantom{x}}$  denotes estimated parameters.